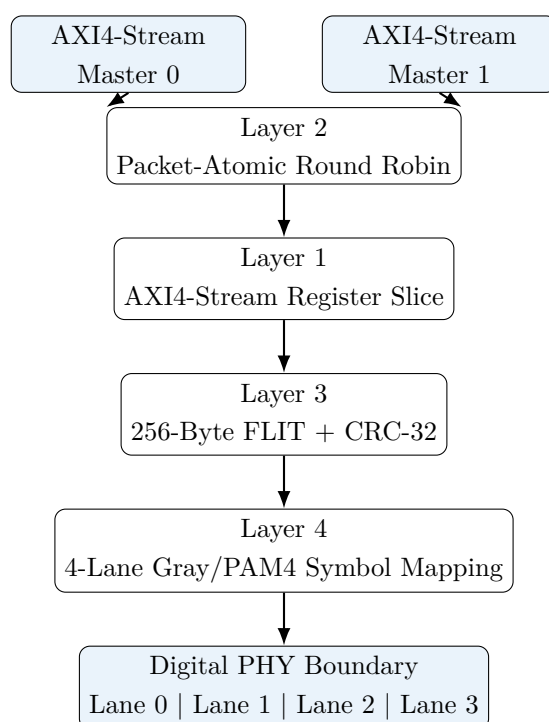


AXI4-Stream to PCIe 6 Link

Four-Layer RTL Architecture and Functional Verification

Stage 1 Technical Submission Report

Semiconductor Challenge



Final functional verification: 162 / 162 checks passed

Layer 2 atomicity invariant: 0 violations

Simulation environment: Icarus Verilog

Prepared from the final RTL, testbenches, simulation logs, and waveform evidence included with the Stage 1 project submission.

Contents

1	Executive Summary	3
2	Problem Statement and Stage 1 Scope	3
2.1	Selected Implementation Scope	4
3	Architecture and Data-Width Budget	4
3.1	Width and Framing Relationships	4
3.2	Architectural Layer Numbering Versus Physical Datapath Order	5
4	Layer 1 - AXI4-Stream Ingress Register Slice	6
4.1	Role	6
4.2	Handshake Mechanism	6
4.3	Example	7
4.4	Backpressure Behavior	7
4.5	Verification Evidence	7
5	Layer 2 - Interconnect and Packet-Atomic Arbiter	8
5.1	Role	8
5.2	Arbitration Policy	8
5.3	Packet Atomicity	8
5.4	Fairness	9
5.5	Verification	9
6	Layer 3 - Link Layer FLIT Packetizer and CRC-32	9
6.1	Role	9
6.2	FLIT Formation	10
6.3	TKEEP Semantics	10
6.4	Early TLAST and Partial FLITs	10
6.5	CRC-32	10
6.6	Verification	11
7	Layer 4 - PCIe Protocol-to-PHY Digital Boundary	11
7.1	Role	11
7.2	Symbolization	11
7.3	Four-Lane Striping	12
7.4	Gray Coding	12
7.5	Verification	12
8	Top-Level Four-Layer Integration	13
8.1	Integrated Datapath	13
8.2	End-to-End Example	13
8.3	Top-Level Verification	14
8.4	Integration Interpretation	14
9	Verification Methodology	14

9.1	Directed Verification	15
9.2	Invariant Monitoring	15
9.3	Scoreboarding and Reference Models	15
9.4	Reset and Backpressure	15
10	Final Verification Results	16
10.1	Reproducibility	16
11	Synthesis and Implementation Observations	16
12	Engineering Trade-Offs	17
12.1	Two-Master Interconnect	17
12.2	One-Beat Register Slice	17
12.3	Fixed 256-Byte FLIT	17
12.4	Combinational CRC and Lane Mapping	17
12.5	Digital PHY Boundary	17
13	Limitations and Future Work	17
14	Conclusion	18
15	Appendix A - Verification Matrix	19
16	Appendix B - Representative Simulation Log Excerpts	19
17	Appendix C - Project Reproducibility	20

1 Executive Summary

The design implements a modular digital datapath corresponding to the four representative layers in the supplied SARCAthon AXI4/AXI4-Stream to PCIe 6 Link architecture. Two independent 512-bit AXI4-Stream sources feed a shared packet-atomic round-robin arbiter. The selected stream passes through a one-beat register slice, is accumulated into 256-byte Flow Information Units (FLITs) with CRC-32 error detection, and is finally transformed into four lanes of 2-bit Gray-coded symbols suitable for a downstream PAM4-oriented SerDes boundary.

The Stage 1 implementation focuses on functional correctness, clean interfaces, independent module verification, integrated verification, and an explicit statement of implementation scope. This is consistent with the supplied Stage 1 objective, which emphasizes correct and verifiable RTL, meaningful functional blocks, integration where applicable, reset/error-condition verification, and simulation evidence. The challenge examples motivate the architecture for streaming packet processing and accelerator data movement.

Verification	162 / 162 checks passed across five final test suites; 0 Layer 2 packet-atomicity violations.
---------------------	---

The design in this project focuses on the digital part of the protocol-to-PHY interface. We have implemented and verified the main RTL functions required for the Stage 1 design, but a complete PCIe controller and the actual physical SerDes hardware are outside the scope of this project. Features such as FEC, analog PAM4 signal generation, and physical-layer implementation can be considered as future work.

2 Problem Statement and Stage 1 Scope

The challenge requires us to design a synthesizable RTL implementation based on the given AXI4/AXI4-Stream to PCIe 6 Link architecture. The main idea is to support high-speed data transfer between different processing blocks and the PCIe interface. The architecture can be useful for applications such as network packet processing, where continuous packet data needs to be transferred efficiently, and machine-learning accelerators, where large amounts of data such as weights, activations, or image data need to be moved between the accelerator and host memory.

The reference architecture identifies four representative layers:

1. AXI4 / AXI4-Stream interface for transaction/data ingress and egress.
2. Interconnect layer for routing, crossbar operation, and arbitration.
3. Link layer for packetization, framing, and CRC/error-handling concepts.
4. PCIe protocol-to-PHY layer for protocol adaptation and lane-oriented PHY-facing data handling.

The Stage 1 requirements focused on developing synthesizable RTL, implementing and integrating the main functional blocks, and verifying the design using directed and self-checking testbenches. The design was also tested for reset and important corner cases, with simulation waveforms and results provided as evidence.

2.1 Selected Implementation Scope

The project selects one concrete and meaningful function from each reference layer, connecting them into an end-to-end flow:

Table 1. Reference architecture to implemented RTL mapping.

Layer	Implemented function	RTL module
AXI4 / AXI4-Stream Interconnect	512-bit streaming ingress and one-beat register slice	<code>layer1_axis_stream_ingress.v</code>
	Two-master packet-atomic round-robin arbitration	<code>layer2_axis_arbiter.v</code>
Link	256-byte FLIT accumulation, TKEEP-aware zero padding, CRC-32	<code>layer3_flit_packetizer.v</code>
Protocol-to-PHY	2-bit symbol generation, four-lane striping, Gray coding	<code>layer4_flit_to_phy_lanes.v</code>
Integration	Complete four-layer connection	<code>top_4layer_true_split.v</code>

3 Architecture and Data-Width Budget

The top-level architecture is shown below. Layer 2 creates a single logical stream, Layer 1 introduces an elastic timing boundary, Layer 3 creates link-layer FLITs, and Layer 4 maps the resulting FLIT-plus-CRC payload to four parallel digital lane outputs.

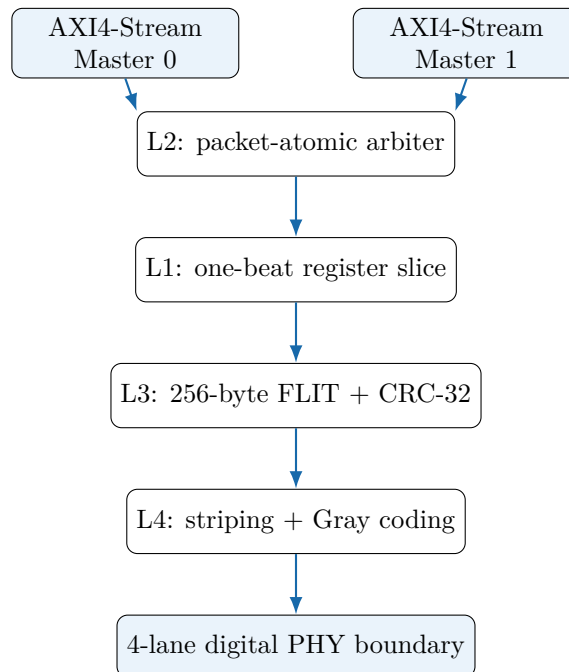


Figure 1. Implemented four-layer datapath.

3.1 Width and Framing Relationships

The design uses a 512-bit AXI4-Stream datapath. Therefore each input beat contains:

$$512 \text{ bits} = \frac{512}{8} = 64 \text{ bytes}$$

The target FLIT size is 256 bytes, giving:

$$\frac{256 \text{ bytes}}{64 \text{ bytes/beat}} = 4 \text{ beats/FLIT}$$

A complete FLIT therefore contains 2048 data bits. With a 32-bit CRC:

$$2048 + 32 = 2080 \text{ bits total}$$

Layer 4 treats these 2080 bits as 2-bit symbols:

$$\frac{2080}{2} = 1040 \text{ symbols}$$

With four lanes:

$$\frac{1040}{4} = 260 \text{ symbols/lane} = 520 \text{ bits/lane}$$

Table 2. Key data-width relationships.

Parameter	Value
AXI data width	512 bits
AXI byte lanes / TKEEP width	64 bits
AXI beat size	64 bytes
FLIT data size	256 bytes = 2048 bits
CRC size	32 bits
FLIT + CRC	2080 bits
Symbols at 2 bits/symbol	1040
Lanes	4
Symbols per lane	260
Bits per lane	520

3.2 Architectural Layer Numbering Versus Physical Datapath Order

The reference architecture defined four functional layers, but the layer numbers do not require the RTL modules to appear in numerical order in the physical pipeline. In this implementation, the two AXI4-Stream sources first enter Layer 2, the packet-atomic arbiter, and are reduced to one logical shared stream. That selected stream then passes through Layer 1, the one-beat ingress register slice, before reaching Layer 3 and Layer 4.

We have used this ordering deliberately. Placing one shared register slice after arbitration provides buffering on the single downstream stream instead of replicating the same register stage separately for each input master. It keeps the Stage 1 implementation compact while preserving the required valid/ready behavior and packet atomicity. The trade-off is that the combinational path from the input masters through the arbiter to the first register boundary remains a timing-closure consideration for a future high-frequency implementation. It does not prevent RTL synthesis; it is an implementation-level optimization consideration for a later stage.

The resulting physical datapath is therefore:

AXI4-Stream Master 0 / Master 1 → Layer 2 Arbiter → Layer 1 Register Slice → Layer 3 FLIT/CRC → Layer 4 Lane Mapping.

This distinction is important when reading the design: the labels “Layer 1” through “Layer 4” describe the functional architecture categories, while the arrows describe the actual RTL connectivity.

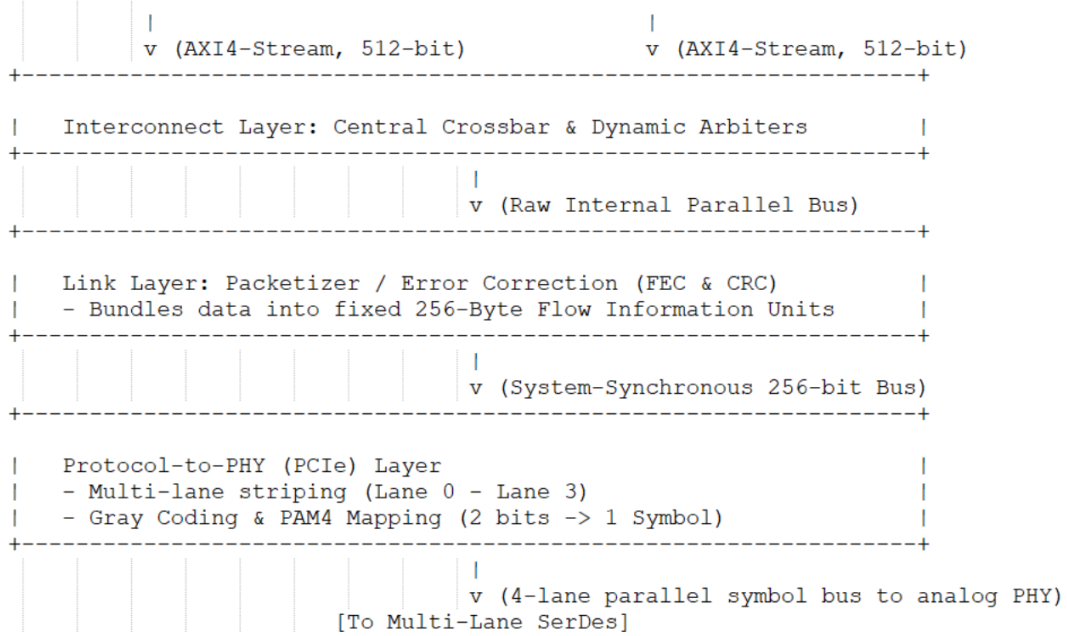


Figure 2. Supplied challenge reference architecture. The submission implements a scoped digital realization of the four representative layers: AXI4/AXI4-Stream, interconnect/arbitration, link-layer packetization/error handling, and protocol-to-PHY lane processing.

4 Layer 1 - AXI4-Stream Ingress Register Slice

4.1 Role

Layer 1 is implemented in `layer1_axis_stream_ingress.v`. It sits between the arbiter and packetizer and provides a one-beat elastic register boundary. The layer does not reinterpret packet boundaries; it preserves data and control information for downstream processing.

The interface consists of 512-bit TDATA, 64-bit TKEEP, one-bit TVALID, TREADY, and TLAST on both sides.

4.2 Handshake Mechanism

The central ready relationship is:

$$s_tready = m_tready \vee \neg m_tvalid$$

In words, the input can accept a new beat when the output is being consumed or when the output register is empty. When the output register is full and the consumer is stalled, the input side is backpressured.

This creates the intended elastic-buffer behavior:

- When downstream is ready, the stage can operate at one beat per cycle.

- When downstream is stalled, the current output transaction is held stable.
- TDATA, TKEEP, and TLAST are transferred together.

4.3 Example

For an ordered input sequence:

```
Beat 0 = 0x...0000
Beat 1 = 0x...0001
Beat 2 = 0x...0002
Beat 3 = 0x...0003, TLAST=1
```

The output preserves the same sequence and packet boundary. The value of adding the register boundary is not transformation; it is controlled buffering and timing isolation.

4.4 Backpressure Behavior

During a stall:

$$m_tvalid = 1, \quad m_tready = 0$$

The valid output transaction remains presented until accepted. This is the key safety property for preventing overwriting of a buffered AXI4-Stream beat.

4.5 Verification Evidence

The final Layer 1 suite reports:

Layer 1

32 / 32 checks passed.

c The checks cover normal in-order traffic, back-to-back transfer behavior, downstream backpressure, reset behavior, and preservation of packet control information.

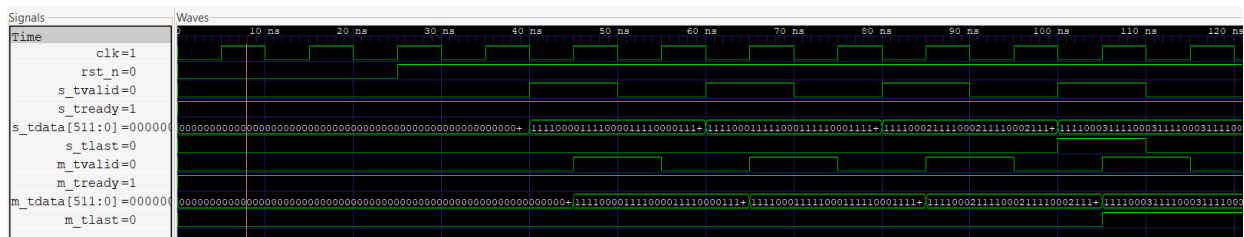


Figure 3. Layer 1 normal AXI4-Stream transfer. Four consecutive 512-bit beats are presented and forwarded in order, with the packet-ending TLAST preserved.

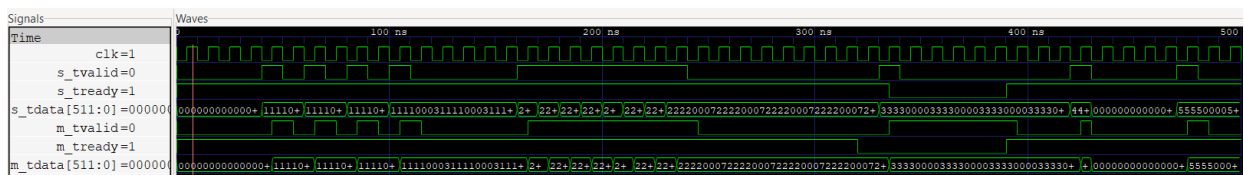


Figure 4. Layer 1 backpressure behavior. The downstream ready signal is removed while a transaction is valid; the registered output remains held until the transfer can complete.

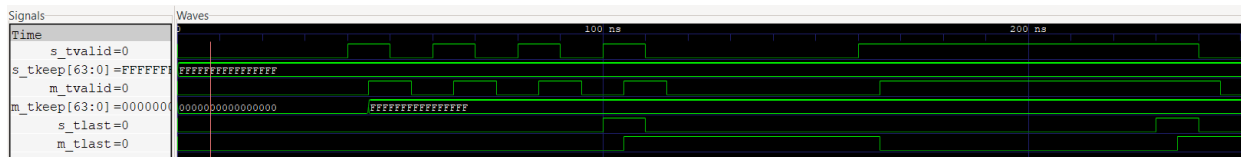


Figure 5. Layer 1 control-field preservation. The waveform shows the relationship between input and output TKEEP/TLAST information across the register boundary.

5 Layer 2 - Interconnect and Packet-Atomic Arbiter

5.1 Role

Layer 2 is implemented in `layer2_axis_arbiter.v`. It accepts two independent AXI4-Stream sources and produces a single downstream stream.

The design intentionally uses a two-master implementation because the goal is to demonstrate the core mechanism - contention, fairness, and packet atomicity - with a compact and thoroughly verified block rather than an unverified generalized N-master crossbar.

5.2 Arbitration Policy

For each master:

$$want0 = s_axis0_tvalid, \quad want1 = s_axis1_tvalid$$

The arbitration cases are:

- Only Master 0 requests: Master 0 is selected.
- Only Master 1 requests: Master 1 is selected.
- Neither requests: the output is idle.
- Both request: round-robin state selects the master that was not served most recently.

5.3 Packet Atomicity

The most important design rule is that a grant is held for the complete packet.

Suppose Master 0 starts a packet:

```
M0 beat 0
M0 beat 1
M0 beat 2
M0 beat 3 + TLAST
```

Master 1 may be requesting in parallel, but it is not granted until the packet-ending transfer occurs. The packetizer behind the arbiter sees one logical stream and therefore relies on this guarantee.

If beat-level switching were allowed, the downstream FLIT construction could observe an invalid sequence such as:

```
M0 beat 0
M0 beat 1
```

```
M1 beat 0 <-- illegal interleave
M0 beat 2 <-- illegal interleave
```

The implementation therefore uses a registered lock/grant state which remains active through the selected master's packet until TLAST is transferred.

5.4 Fairness

When both masters are continuously active, the next winner alternates according to the identity of the master most recently served. This prevents persistent preference for one source when both are requesting the shared resource.

5.5 Verification

The Layer 2 testbench contains directed arbitration scenarios, contention traffic, packet boundary checks, a packet-atomicity invariant monitor, and source-attribution checking.

Layer 2

33 / 33 checks passed; 0 atomicity invariant violations.

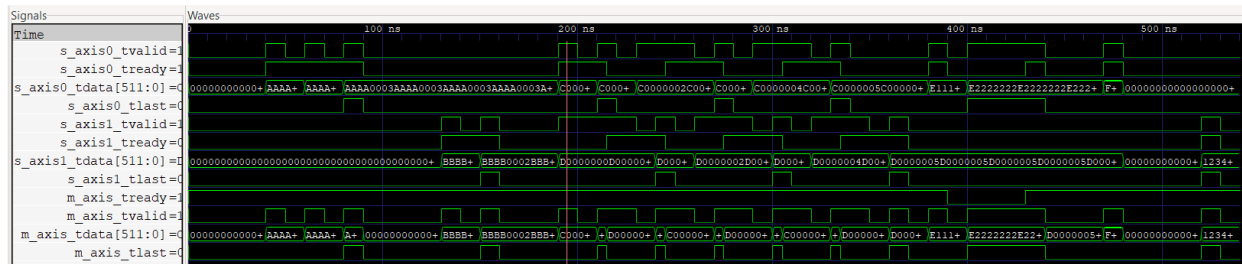


Figure 6. Layer 2 two-master contention. Both sources generate traffic while the arbiter controls a single output channel. The waveform shows the valid/ready interactions and packet boundaries involved in arbitration.

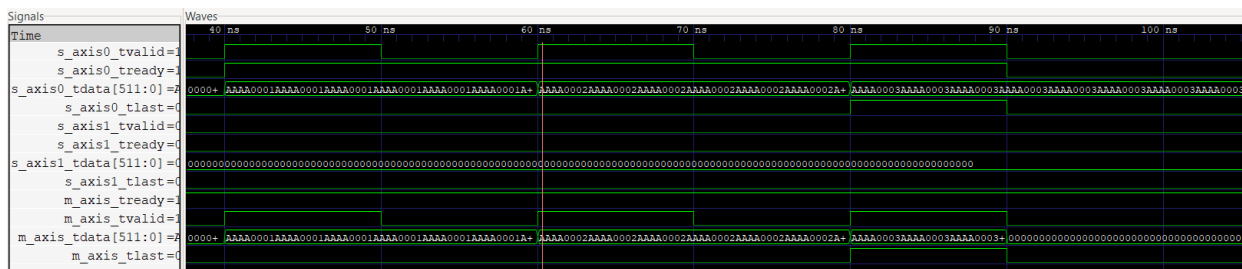


Figure 7. Layer 2 packet atomicity zoom. The selected master retains the channel across multiple beats while the competing master is prevented from transferring until the packet-ending TLAST.

Verification property: once a packet starts, the granted source must not change before the selected packet's TLAST transfer. The final monitor reported **0 violations**.

6 Layer 3 - Link Layer FLIT Packetizer and CRC-32

6.1 Role

Layer 3 is implemented in `layer3_flit_packetizer.v`. It converts the 512-bit stream into fixed-size 256-byte FLITs and computes an associated CRC-32 value.

6.2 FLIT Formation

Because each input beat is 64 bytes, exactly four beats fill one complete FLIT:

$$4 \times 512 = 2048 \text{ bits} = 256 \text{ bytes}$$

The accumulation buffer therefore holds 2048 data bits.

The packetizer has two principal states:

1. **ACCUM**: accept and store incoming beats at the current word index.
2. **PRESENT**: hold the completed FLIT and CRC until the downstream stage accepts them.

6.3 TKEEP Semantics

The 64-bit **TKEEP** corresponds one-to-one with the 64 bytes of a 512-bit input beat.

For byte position i :

$$TKEEP[i] = 1 \Rightarrow \text{copy byte } i$$

$$TKEEP[i] = 0 \Rightarrow \text{write byte } i \text{ as } 0$$

The bytes are not compacted. This preserves byte positions inside the FLIT and makes partial final beats deterministic.

6.4 Early TLAST and Partial FLITs

A packet may terminate before four beats have arrived. If **TLAST** is asserted early, the packetizer flushes the accumulated content, zero-pads the unused region, marks the result using `m_flit_partial`, and computes CRC over the resulting 256-byte representation.

For example, a two-beat packet produces:

```
Beat 0 -> 64 bytes
Beat 1 -> 64 bytes + TLAST
Remaining 128 bytes -> zero padding
Total FLIT representation -> 256 bytes
```

6.5 CRC-32

The design uses the IEEE 802.3 CRC-32 polynomial:

$$0x04C11DB7$$

The implementation uses the reflected/LSB-first form for its byte update procedure. The CRC is generated over the 256-byte FLIT representation, including any zero padding required to complete the fixed-size structure.

A complete output therefore has:

$$2048 \text{ data bits} + 32 \text{ CRC bits} = 2080 \text{ bits}$$

6.6 Verification

Layer 3

29 / 29 checks passed.

The verification covers complete FLIT formation, partial packets, early TLAST, TKEEP-aware byte masking, zero padding, CRC generation, and output handshake behavior.

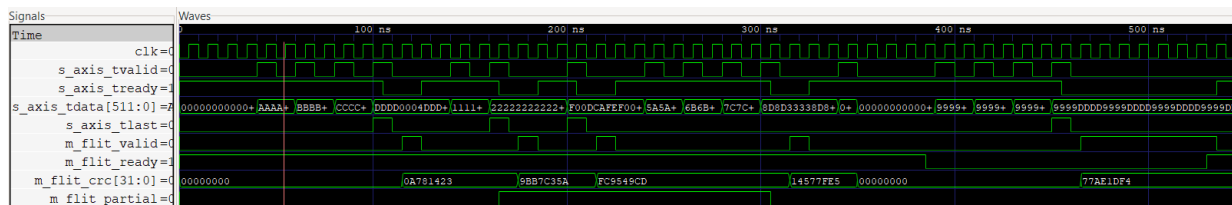


Figure 8. Layer 3 FLIT formation overview. The input AXI4-Stream beats are accumulated until a complete FLIT or early packet termination causes the FLIT to be presented with its CRC.

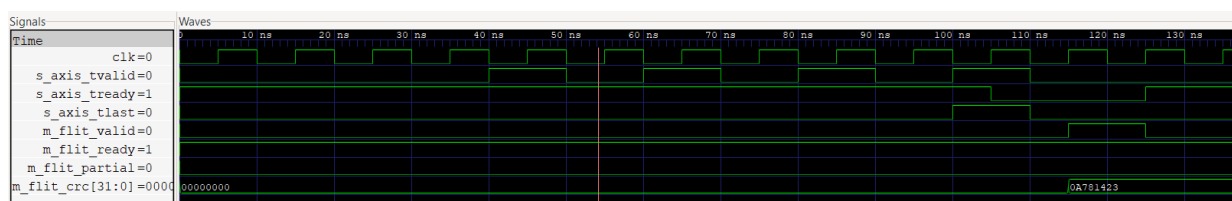


Figure 9. Layer 3 early packet termination. The input packet ends before four beats, causing a partial FLIT indication and corresponding CRC output.

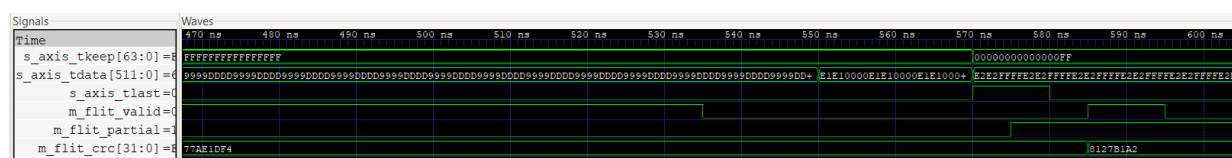


Figure 10. Layer 3 TKEEP-aware partial FLIT generation. A non-uniform TKEEP value is exercised on the final input beat, with packet termination and the resulting partial FLIT/CRC event visible in the waveform.

7 Layer 4 - PCIe Protocol-to-PHY Digital Boundary

7.1 Role

Layer 4 is implemented in `layer4_flit_to_phy_lanes.v`. It accepts the 2080-bit FLIT-plus-CRC payload and converts it into four parallel lane words representing the digital symbol streams that a downstream PAM4 SerDes could consume.

The module intentionally stops at the digital boundary. It does not model analog voltage generation, serializer circuitry, equalization, clock recovery, or the physical channel.

7.2 Symbolization

The input contains 2080 bits. With two bits per PAM4 symbol:

$$2080/2 = 1040 \text{ symbols}$$

7.3 Four-Lane Striping

Symbol index i maps to:

$$lane = i \bmod 4$$

with its position inside the lane given by:

$$position = \lfloor i/4 \rfloor$$

This creates exactly 260 symbols per lane:

$$1040/4 = 260$$

Since each symbol contains two bits:

$$260 \times 2 = 520 \text{ bits per lane}$$

Thus the four outputs are 520-bit parallel lane buses.

7.4 Gray Coding

Each 2-bit symbol is mapped using:

Table 3. 2-bit Gray mapping used by Layer 4.

Raw symbol	Gray-coded symbol
00	00
01	01
10	11
11	10

This mapping makes adjacent symbol levels differ in only one bit, which is the relevant digital coding property when these codes are later interpreted as adjacent PAM4 signal levels.

7.5 Verification

Layer 4

39 / 39 checks passed.

The testbench covers symbol generation, deterministic lane striping, Gray coding, output handshaking, backpressure, and reset recovery.

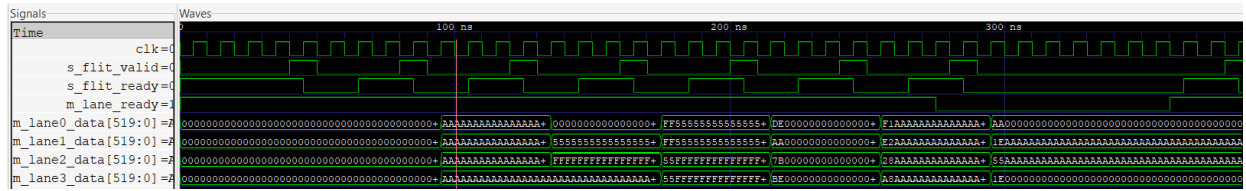


Figure 11. Layer 4 lane stripping. The FLIT-plus-CRC payload is transformed into four deterministic lane streams, with all four lane outputs shown together.

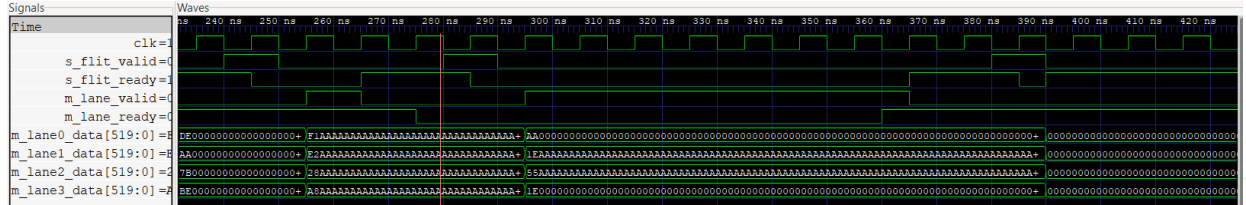


Figure 12. Layer 4 lane-side backpressure. While the PHY-side consumer is not ready, the valid lane bundle remains presented until the transfer can complete.

8 Top-Level Four-Layer Integration

8.1 Integrated Datapath

The complete system is instantiated in `top_4layer_true_split.v`. Both AXI4-Stream masters enter the shared Layer 2 arbiter. The selected stream then flows through Layer 1, Layer 3, and Layer 4 without any special-case bypass.

The ordering is intentional: Layer 2 first creates a single logical stream, then Layer 1 inserts a register boundary on that shared path. The functional layer numbering follows the challenge categories rather than imposing a mandatory numerical pipeline order.

This separation is an important design feature: each module has a clear contract and can be verified independently before being exercised in the combined datapath.

8.2 End-to-End Example

Consider a four-beat packet generated by Master 0:

```
M0 Beat 0
M0 Beat 1
M0 Beat 2
M0 Beat 3 + TLAST
```

The end-to-end operation is:

1. Layer 2 grants Master 0 and holds ownership through TLAST.
2. Layer 1 buffers and forwards the 512-bit beats.
3. Layer 3 accumulates four beats to form a 256-byte FLIT and generates CRC-32.
4. Layer 4 receives the 2080-bit FLIT-plus-CRC representation.
5. Layer 4 produces 1040 two-bit symbols and maps them across four lanes.

6. Each lane therefore contains 260 symbols, represented as a 520-bit output word.

8.3 Top-Level Verification

The final integration testbench contains multiple-master traffic, packetized output checking, CRC/lane reference checking, and a downstream backpressure scenario.

The saved result is:

```
FLITs sent (scoreboard pushes): 9
FLITs lane-checked (main scenario): 6
Grant counts: master0=7 master1=2
TESTS PASSED: 29 TESTS FAILED: 0
RESULT: ALL TESTS PASSED
```

Integration

29 / 29 checks passed.

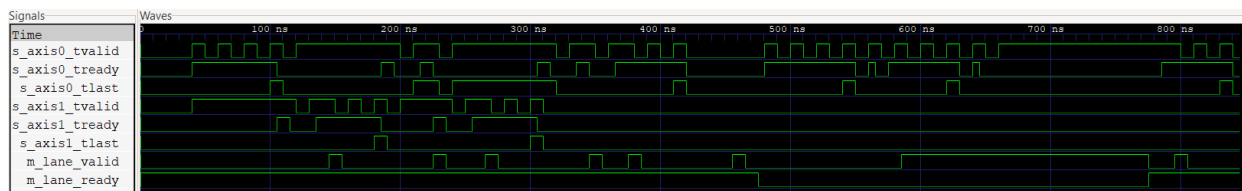


Figure 13. Top-level end-to-end transfer. Two independent AXI4-Stream sources feed the integrated pipeline and ultimately drive the four-lane PHY-facing output handshake.

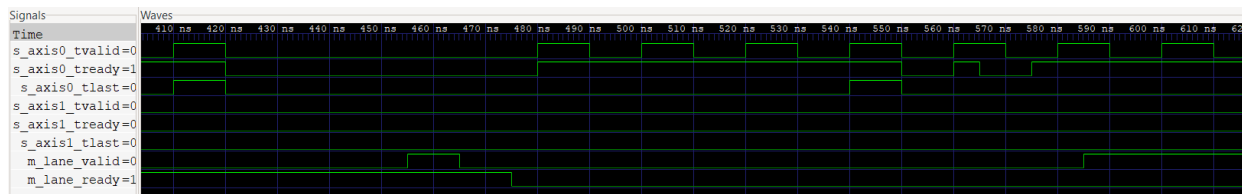


Figure 14. Top-level integrated backpressure. The PHY-facing ready signal is removed during active traffic, exercising the valid/ready behavior of the complete datapath.

8.4 Integration Interpretation

The top-level waveform mainly shows the overall behavior of the complete design, so all the internal handshake signals are not visible at the same time. Therefore, we have used separate waveforms for each layer to show the important details of that layer. For example, the Layer 1 and Layer 4 waveforms show how the data is held during backpressure, while the top-level waveform is used to show the overall response of the integrated design and its end-to-end operation.

9 Verification Methodology

The verification strategy is divided into two levels. First, each module is tested separately using directed and self-checking test cases to make sure that the individual functions are working correctly. After that, all the modules are connected in the top-level design and tested together to make sure that the complete data flow works correctly from the input AXI4-Stream interfaces to the final output lanes.

9.1 Directed Verification

Directed tests were used where the expected result is deterministic and interpretable. Examples include:

- basic in-order AXI transfers;
- continuous back-to-back transfers;
- downstream stalls and recovery;
- simultaneous master contention;
- packet boundaries and TLAST;
- partial FLIT creation;
- non-uniform TKEEP;
- deterministic lane patterns; and
- reset behavior.

9.2 Invariant Monitoring

Layer 2 includes a runtime atomicity monitor. The monitor observes packet ownership while the arbiter is locked and reports a violation if the other master becomes transferable before the selected packet's TLAST.

The final run reports:

Atomicity	0 invariant violations.
------------------	-------------------------

9.3 Scoreboarding and Reference Models

The verification environment uses fixed expected values for the test cases where the output is predictable. In the Layer 2 contention test, different data patterns are used for each master so that the output can be checked against the correct source. For Layer 3 and the complete integrated design, a separate CRC reference is used to compare the calculated CRC with the DUT output.

The use of this independent table-driven CRC reference in the verification environment reduces the risk of reproducing the same procedural mistake in both the DUT and checker.

9.4 Reset and Backpressure

Reset is included as an important part of the design verification. Each module is tested during reset and after reset to make sure that no old or incomplete transaction is left in the design. The testbenches also check that the modules return to a known idle state and can correctly process new data after reset.

Backpressure is tested at both individual module level and in the complete design. When the receiving side is not ready, the current valid transaction is held stable until the receiver can accept it. This helps ensure that data is not lost or changed when the pipeline is stalled.

10 Final Verification Results

The final Stage 1 project was run from the submission's Makefile using Icarus Verilog. The five final suites reported no functional failures.

Table 4. Final verification summary.

Verification suite	Checks	Failed	Result
Layer 1 - AXI4-Stream ingress	32	0	PASS
Layer 2 - arbitration/atomicity	33	0	PASS ; atomicity violations = 0
Layer 3 - FLIT/CRC	29	0	PASS
Layer 4 - lane mapping	39	0	PASS
Full four-layer integration	29	0	PASS
Total	162	0	162/162 PASS

The corresponding saved simulation result logs are included in the project under **results/**. Each layer also produces a VCD waveform, allowing the reported behaviors to be independently inspected.

10.1 Reproducibility

The project Makefile provides one-command regression execution:

```
make all
```

Individual suites can also be run independently:

```
make layer1
make layer2
make layer3
make layer4
make top
```

The project includes the generated logs and VCD files as evidence from the final code state.

11 Synthesis and Implementation Observations

The project also includes a basic Yosys check to make sure that the RTL can be processed correctly at a structural level. This was used only as an initial check of the design and is not considered a complete synthesis or Stage 2 implementation result.

Table 5. Preliminary Yosys observations included with the submission.

Module	Observation
Layer 1	Clean structural analysis; no inferred latches reported.
Layer 2	Clean structural analysis; no inferred latches reported.
Layer 3	Elaboration begins, but optimization/checking did not complete within the allotted tool budget.
Layer 4	Verilog frontend parsing did not complete within the allotted tool budget.

The large combinational footprint is explainable from the architecture: Layer 3 computes CRC over a full 256-byte FLIT in a single combinational construction, while Layer 4 performs the symbol/lane transformation over the full 1040-symbol representation. These structures are suitable for functional Stage 1 verification but are natural candidates for pipelining and timing-aware decomposition in a future implementation stage.

The result is mentioned clearly in the report so that the limitations of the current implementation are also explained. Passing the simulation tests shows that the design is functionally working, while synthesis and timing optimization are separate issues that can be improved in future stages.

12 Engineering Trade-Offs

12.1 Two-Master Interconnect

A two-master arbiter is enough for this project to show the main functions of the interconnect layer, such as handling contention, maintaining packet ownership, and providing round-robin fairness. Keeping the design limited to two masters also makes the RTL simpler and allows us to verify its behavior more thoroughly.

12.2 One-Beat Register Slice

The register slice adds a small amount of latency while providing timing isolation and controlled buffering. The benefit is a clean valid/ready boundary without requiring a deeper queue.

12.3 Fixed 256-Byte FLIT

With a 512-bit input width, a 256-byte FLIT maps exactly to four beats. This removes a full-size packing remainder and makes the packetizer easy to reason about while still allowing early packet termination.

12.4 Combinational CRC and Lane Mapping

The current design focuses on keeping the functionality simple and easy to understand. The main trade-off is that some operations use a larger amount of combinational logic. This may affect timing and area in a more advanced implementation, so these parts can be optimized and pipelined in future work.

12.5 Digital PHY Boundary

The PHY-facing interface is represented as four parallel digital lane words. This keeps analog circuitry and physical-layer implementation outside the RTL while still implementing the protocol-to-PHY digital functions named in the challenge architecture.

13 Limitations and Future Work

The following features are not included in the current Stage 1 implementation and are left for future work:

- a complete N-master crossbar;
- complete PCIe transaction, data-link, and configuration protocols;

- forward-error correction beyond CRC-32 error detection;
- analog PAM4 voltage generation;
- physical SerDes serialization and receiver recovery;
- timing closure and physical implementation; or
- full GDSII generation.

These features can be taken up in future stages as the design is further developed beyond the current Stage 1 implementation.

Future work can include:

1. generalized multi-master arbitration and routing;
2. deeper elastic buffering and higher-throughput flow control;
3. pipelined or byte-parallel CRC computation;
4. formal protocol assertions;
5. constrained-random verification;
6. full PCIe protocol-layer behavior;
7. FEC integration;
8. actual SerDes/PHY modeling;
9. synthesis, timing, power, and area optimization; and
10. FPGA or ASIC implementation and physical-design validation.

14 Conclusion

This project demonstrates a modular and functionally verified digital realization of the four principal layers represented in the SARCathon Stage 1 reference architecture.

The design accepts two independent 512-bit AXI4-Stream sources, arbitrates them using packet-atomic round-robin scheduling, introduces a registered streaming boundary, forms 256-byte FLITs with CRC-32 protection, and maps the resulting 2080-bit representation into four lanes of 2-bit Gray-coded symbols.

The strongest architectural property is the clean separation of responsibilities: arbitration guarantees logical packet ownership, Layer 1 guarantees streaming elasticity, Layer 3 defines deterministic link-layer framing, and Layer 4 defines the digital lane representation at the PHY boundary.

The final simulation campaign reports:

162 / 162 checks passed
0 packet-atomicity invariant violations

The result is therefore a compact, modular, reproducible Stage 1 RTL implementation supported by module-level verification, integrated verification, saved simulation logs, and waveform evidence.

15 Appendix A - Verification Matrix

Table 6. Detailed verification focus by layer.

Layer	Verification focus	Count	Status
Layer 1	Basic transfer, continuous burst, backpressure, stable output, reset, packet controls	32	PASS
Layer 2	Arbitration cases, contention, packet framing, fairness behavior, source attribution, atomicity monitor	33	PASS
Layer 3	Full FLIT, partial FLIT, early TLAST, TKEEP masking, zero padding, CRC, backpressure	29	PASS
Layer 4	Symbol generation, four-lane striping, Gray mapping, output handshake, backpressure	39	PASS
Integration	Multi-master traffic, end-to-end FLIT/lane checking, integrated backpressure	29	PASS
Total		162	PASS

16 Appendix B - Representative Simulation Log Excerpts

Layer 1

```
TESTS PASSED: 32 TESTS FAILED: 0
RESULT: ALL TESTS PASSED
```

Layer 2

```
TESTS PASSED: 33 TESTS FAILED: 0
Atomicity invariant violations: 0
RESULT: ALL TESTS PASSED
```

Layer 3

```
TESTS PASSED: 29 TESTS FAILED: 0
RESULT: ALL TESTS PASSED
```

Layer 4

```
TESTS PASSED: 39 TESTS FAILED: 0
RESULT: ALL TESTS PASSED
```

Full integration

```
FLITs sent (scoreboard pushes): 9
```

```
FLITs lane-checked (main scenario): 6
Grant counts: master0=7 master1=2
TESTS PASSED: 29 TESTS FAILED: 0
RESULT: ALL TESTS PASSED
```

17 Appendix C - Project Reproducibility

The final project contains the following major elements:

- `rtl/`: five RTL modules including the four layers and top-level integration;
- `tb/`: five corresponding verification environments;
- `results/`: final simulation logs and generated VCD waveforms;
- `synthesis/`: preliminary Yosys structural observations;
- `physical_design/`: Stage 1 scope note for later implementation;
- `reports/`: this technical design document;
- `Makefile`: reproducible compilation and simulation targets; and
- `README.md`: project structure and execution instructions.